STUDENT FEEDBACK MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

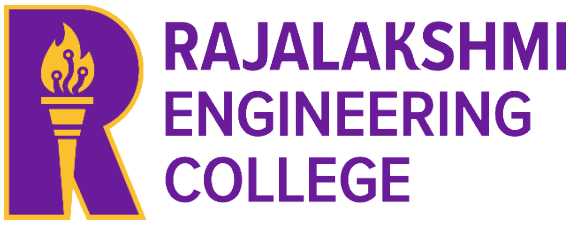
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled “STUDENT FEEDBACK MANAGEMENT SYSTEM” is the bonafide work of YOGESHWARI P (2116241501254), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year [2026–2027], Semester [V], under my supervision.

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DECLARATION

I, YOGESHWARI P(2116241501254) hereby declare that the project report titled “Student Feedback Management System”,submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
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I also acknowledge [Zoho Corporation / Zoho Creator Education Programme] for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

YOGESHWARI P

ABSTRACT

The Student Feedback Management System is a low-code application developed using Zoho Creator to simplify the collection, management, and analysis of academic feedback in educational institutions. In many colleges, student feedback is collected through paper-based questionnaires, spreadsheets, or separate online forms. These conventional methods often result in scattered information, duplicate records, manual calculation errors, difficulties in maintaining semester-wise feedback, and increased administrative effort. Identifying areas that require improvement in course content and teaching effectiveness also becomes challenging when feedback information is not maintained in a centralized system. To overcome these limitations, a centralized digital solution was developed to streamline feedback collection, organize academic records, automate routine operations, and provide a clear overview of course evaluation results.

The application was developed using the low-code capabilities of Zoho Creator, enabling the creation of functional application components with minimal programming effort. The system consists of four interconnected modules: Student, Course, Feedback Response, and Semester Summary. The Feedback Response module collects student evaluations based on Content Quality, Delivery, Clarity, and Engagement, along with Overall Satisfaction, Suggestions, and an anonymity preference. Lookup relationships connect student and course information with the corresponding feedback records, improving data organization and consistency. Eight reports, including All Students, All Courses, All Responses, All Semester Summaries, Course Types, Responses Board, Student Statuses, and Trend Statuses, provide structured access to application information. An interactive dashboard presents key performance indicators, average feedback ratings, total responses, and visual representations of semester performance trends. Deluge scripting and workflow automation were configured to automatically populate student email addresses, validate numerical feedback ratings, and prepare scheduled semester summary reports. The application also includes Blueprint components for Course Feedback Collection and Semester Performance Review.

The application was initially evaluated using sample academic records to examine data entry, lookup relationships, feedback submission, report generation, and dashboard visualization. The implementation demonstrates how a low-code platform can reduce repetitive data management activities, improve the organization of student feedback, and simplify the presentation of academic evaluation information. Additional verification of scheduled workflow execution, Blueprint transitions, and user access permissions is required before wider deployment. The developed system provides a structured foundation for academic feedback management and can be extended with automated summary calculations, role-based access control, advanced analytics, and AI-based sentiment analysis. Overall, the project highlights the practical application of Zoho Creator in developing a centralized, maintainable, and user-friendly educational feedback management solution.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Student Feedback Management System; Course Evaluation; Dashboard Analytics.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Student Feedback Management System
Application Domain	Education
Student Name & Register No.	YOGESHWARI P - 2116241501254
Zoho Creator Edition	Zoho Creator Event Edition (Education)
Application Name (Link Name)	Student_Feedback_Management_System
Published / Live URL	https://creator.zoho.in/appbuilder/liayogeshwari/student-feedback-collection-and-course-evaluat1/edit
Demo Credentials (per role)	Administrator: admin@example.com Student: student@example.com
Number of Forms / Subforms	Forms: 4 Subforms: 1
Number of Reports (by type)	List Reports: 6 Kanban Reports: 1 Calendar Reports: 0 Pivot/Chart Reports: 1 Other Reports: 4
Number of Pages / Dashboards	1
Number of Workflows	Form Workflows: 4 Scheduled Workflows: 1 Approval Workflows: 0 Blueprints: 2
Deluge Functions / Custom Actions	3
Integrations	Zoho Creator Built-in Services Zoho Mail (Email Notifications) Zoho AI Assistant
Roles & Profiles Defined	Administrator Student
Environments Used	Zoho Creator Cloud Environment
Date of Production Deployment	

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LIST OF ABBREVIATION

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Student feedback is an essential component of academic quality improvement in educational institutions. It enables students to express their opinions regarding course content, teaching methods, clarity of instruction, student engagement, and overall learning experience. Traditionally, feedback is collected through paper-based questionnaires, spreadsheets, or basic online forms. However, managing large numbers of responses, maintaining accurate records, and preparing consolidated feedback summaries through these methods can be time-consuming and difficult.

The growing adoption of low-code application development platforms has introduced a more efficient approach to developing customised academic feedback management systems. Zoho Creator provides a visual development environment with built-in facilities for creating forms, reports, dashboards, and automated workflows. It also supports Deluge scripting, which enables developers to implement validation rules and application-specific operations with reduced manual coding. These features make Zoho Creator suitable for developing a structured and accessible student feedback management application.

The Student Feedback Management System has been developed using Zoho Creator to simplify and automate academic feedback collection and evaluation. The application consists of four main modules: Student, Course, Feedback Response, and Semester Summary. It incorporates lookup relationships, feedback rating validation, automated email notifications, reports, dashboard analytics, and Blueprint-based process management. By digitising student feedback management, the application reduces manual effort, improves data organisation, and provides a centralised platform for analysing course evaluation results and supporting academic quality improvement.

1.2 Problem Statement

Educational institutions often collect student feedback using manual methods such as paper-based questionnaires, spreadsheets, and basic online forms. These approaches make it difficult to organise feedback responses, maintain accurate student and course records, calculate evaluation ratings, and generate consolidated reports efficiently.

Pain Points

- Manual collection and processing of student feedback require significant time and administrative effort..
- Feedback records maintained across multiple spreadsheets or documents may result in duplicate entries and inconsistent information.
- Lack of centralised feedback analysis makes it difficult for faculty members and administrators to monitor course evaluation results effectively.
- Manual calculation of feedback ratings and preparation of semester-wise summary reports increase the possibility of human errors and delays.

1.3 Objectives

- To design a structured database and user-friendly forms for managing student details, course information, feedback responses, and semester summaries.
- To automate student feedback management using Deluge scripting, form workflows, scheduled workflows, and Blueprint-based process management.
- To provide reports and an interactive dashboard for monitoring feedback responses, course evaluation ratings, and semester performance.
- To implement feedback rating validation, automated email notifications, and reusable Deluge functions for improving data processing and reducing manual effort.
- To configure role-based access permissions for administrators, students, and academic staff using Zoho Creator's built-in security features.
- To improve the accuracy, accessibility, and organisation of academic feedback through centralised data management and workflow automation.

1.4 Scope of the Project

The scope of the Student Feedback Management System includes the development of a web-based application using Zoho Creator for collecting, managing, and analysing academic feedback within an educational institution. The system supports three primary user roles: Administrator, Student, and Academic Staff. The application includes four main modules: Student, Course, Feedback Response, and Semester Summary. It also incorporates a Course Feedback Criteria subform, reports, an interactive dashboard, Deluge scripting, workflow automation, and Blueprint-based process management.

The application enables administrators to manage student and course information, maintain feedback records, configure workflows, and monitor evaluation results. Students can submit feedback for their respective courses by providing ratings for Content Quality, Delivery, Clarity, and Engagement, along with Overall Satisfaction and Suggestions. Academic staff can access relevant feedback reports and dashboard analytics to review course evaluation results. Automated workflows support feedback validation, email notifications, and scheduled semester summary reporting.

The project focuses primarily on academic feedback collection, course evaluation, data organisation, and report generation. It does not include online examinations, attendance management, fee management, library management, hostel management, or a complete Learning Management System (LMS). Advanced features such as AI-based sentiment analysis, predictive academic analytics, and extensive external API integrations are outside the current implementation scope and may be considered for future enhancement.

1.5 Justification for a Low-Code Approach

Zoho Creator is a suitable platform for developing the Student Feedback Management System because it provides a low-code development environment that reduces application development time while maintaining flexibility and ease of customisation. The platform enables developers to create forms, reports, dashboards, workflows, and application-specific business logic using visual development tools and Deluge scripting instead of extensive traditional programming.

The low-code approach simplifies application maintenance and allows future modifications with reduced development effort. Zoho Creator provides built-in database management, cloud hosting, user authentication, permission configuration, workflow automation, and reporting facilities. These features support the development of a centralised feedback management application without requiring separate infrastructure for each application component.

Compared with traditional software development, Zoho Creator reduces coding complexity, simplifies application configuration, and supports rapid development of academic management solutions. Its visual development environment, reusable components, and automation capabilities make it an effective platform for implementing the Student Feedback Management System while supporting maintainability and future scalability.

1.6 Organisation of the Report

Chapter 2 reviews the existing student feedback collection system, its limitations, related low-code literature, and the proposed system. Chapter 3 presents the requirement analysis, including stakeholders, functional and non-functional requirements, hardware and software requirements, and the use case model. Chapter 4 describes the system architecture, data model, form design, workflow design, user interface, and access control design. Chapter 5 explains the implementation of the Student Feedback Management System using Zoho Creator, including forms, reports,

dashboards, Deluge scripting, workflows, Blueprints, and integrations. Chapter 6 covers testing, validation, debugging, and test results. Chapter 7 discusses application access, deployment status, performance considerations, security, and governance. Chapter 8 presents the application results, achievement of objectives, comparison with the existing system, and limitations of the low-code approach. Finally, Chapter 9 concludes the report with future enhancements and learning outcomes.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In many educational institutions, student feedback is collected using traditional methods such as paper-based questionnaires, spreadsheets, basic online forms, and manual registers. Students provide their opinions regarding course content, teaching effectiveness, clarity of instruction, and overall learning experience. Faculty members or academic administrators collect these responses and manually organise the information for further evaluation.

The collected feedback is generally maintained in separate documents or spreadsheets. Academic staff must review individual responses, calculate average ratings, compare course evaluations, and prepare semester-wise summary reports manually. Since these activities are performed using different tools, maintaining consistency and retrieving historical feedback information can become difficult.

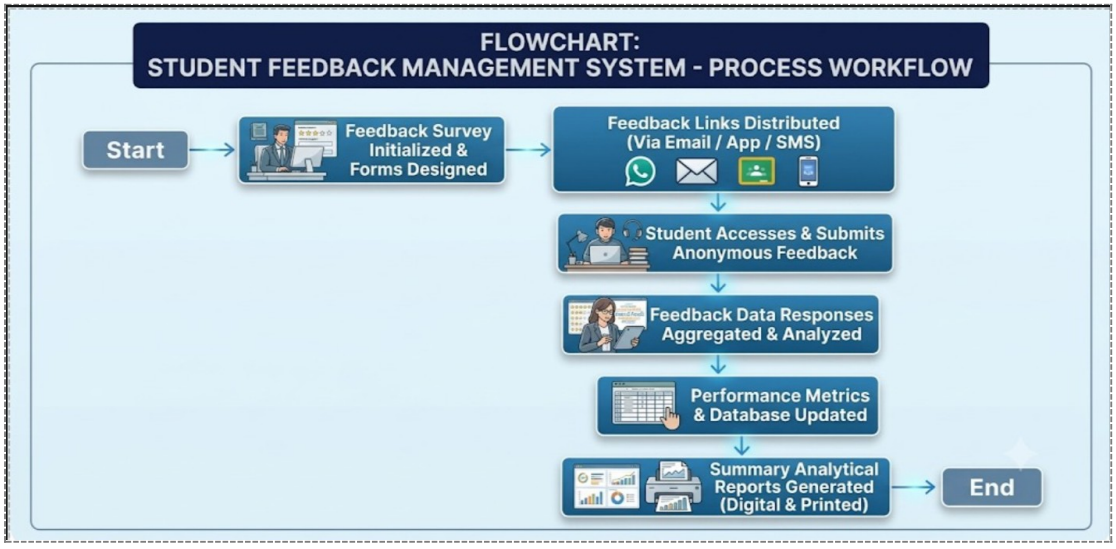


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Student feedback is collected manually through paper forms, spreadsheets, or separate online questionnaires.
- There is no centralized database to maintain student academic activities.
- Faculty members and administrators spend significant time preparing consolidated feedback reports.
- Manual calculation of course evaluation ratings increases the possibility of errors.
- Comparing feedback result across different courses and semesters is difficult.
- Duplicate records and inconsistent information, and delays may occur because feedback data is maintained across multiple sources.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development Time	8–10 Weeks	2–3 Weeks
Skills Required	Advanced programming knowledge	Basic programming and low-code knowledge
Hosting & Infrastructure	Separate server configuration required	Built-in cloud hosting
Database Management	Requires database design and integration	Built-in database with lookup relationships
Security & Access Control	Requires manual configuration and implementation	Built-in authentication and role permissions.
Maintenance & Changes	May require extensive code modifications	Easier modification of forms and workflows
Customisation	Extensive customisation through programming	Visual development with Deluge-based customisation
Workflow Automation	Requires custom programming	Visual workflows and Deluge scripting

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No	Author / Source	Title / Product	Key Findings	Relevance to this Project
1	Zoho Corporation (2024)	Zoho Creator Documentation	Explains forms, reports, workflows, Deluge, Blueprint and application configuration.	Used as the primary development platform.
2	Gartner (2023)	Enterprise Low-Code Application Platforms	Low-code platforms reduce development time and improve productivity.	Supports the selection of Zoho Creator.
3	Zoho Corporation	Deluge Script Reference Guide	Provides information about implementing custom logic and workflow automation.	Supports feedback validation and automated processing.
4	Academic Feedback Management Approaches	Student Feedback Collection Systems	Centralized feedback management improves academic performance and monitoring.	Forms the conceptual basis for the proposed application.
5	Educational Management System Study	Academic Information Management	Digital academic management improves efficiency and reduces manual work.	Supports the need for centralized academic feedback management.

2.5 Emerging Trends in Low-Code Development

Low-code development has become an important approach to building applications because it reduces the amount of manual programming required and simplifies application creation. Modern low-code platforms provide visual development tools, workflow automation, reusable components, cloud hosting, and built-in reporting capabilities to improve development efficiency.

The Student Feedback Management System demonstrates the practical application of these capabilities through Zoho Creator. Forms and lookup relationships organise student, course, and feedback information, while Deluge scripting supports validation and application-specific operations. Automated workflows, email notifications, and Blueprint-based process management help reduce repetitive administrative activities.

2.6 Proposed System

The proposed Student Feedback Management System is a centralised web-based

application developed using Zoho Creator to simplify the collection, management, and analysis of academic feedback. The system consists of four main modules: Student, Course, Feedback Response, and Semester Summary. Lookup relationships connect the relevant records, while the Course Feedback Criteria subform maintains evaluation criteria associated with courses.

The application allows students to submit feedback based on Content Quality, Delivery, Clarity, Engagement, Overall Satisfaction, and Suggestions. Deluge scripts validate feedback ratings, while automated workflows support student confirmation emails, academic staff notifications, and scheduled semester summary reporting. Reports and an interactive dashboard provide structured access to feedback information and course evaluation results.

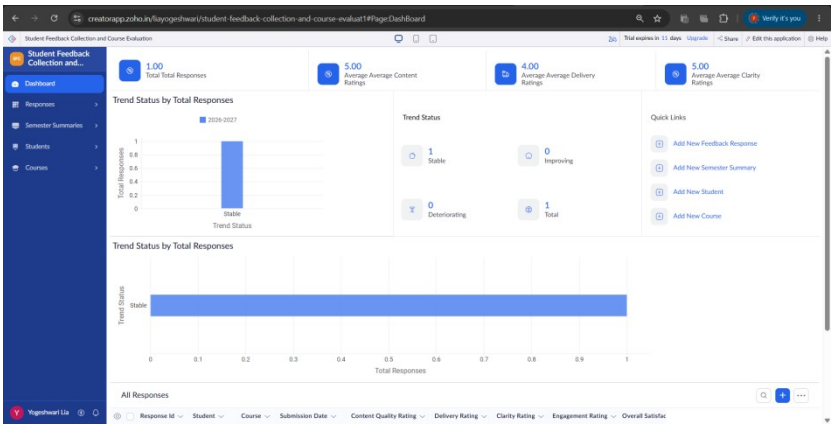


Figure 2.2: Proposed Student Feedback Management System

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	The administrator manages the complete Student Feedback Management System and controls application settings.	Manages student and course records, feedback information, workflows, reports, and user permissions.
Student	Students are the end users who view and update their assigned academic feedback activities.	Views available courses, submits course evaluations, and accesses permitted personal feedback records.
Faculty / Mentor	Faculty members monitor student academic feedback activities and review task progress.	Manages course information and reviews feedback reports, evaluation summaries, and dashboard analytics.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow administrators to create and manage student profiles with required details such as name, register number, department, and semester, email.	Administrator	High
FR-02	The system shall allow administrators to create and manage course details, including course ID,Course Name, Course Code, Instructor, Semester, and Credits.	Administrator	High
FR-03	The system shall allow students to submit course feedback using Content Quality, Delivery, Clarity, Engagement, Overall Satisfaction, and Suggestions.	Student	High
FR-04	The system shall validate feedback ratings before submission using Deluge scripting to ensure accurate data entry.	Administrator, Student	High
FR-05	The system shall generate feedback reports and dashboard analytics to monitor course ratings, total responses, and semester-wise performance trends.	Administrator, Academic Staff	High
FR-06	The system shall support automated notifications and scheduled semester summary reporting through Zoho Creator workflows.	Administrator, Academic Staff	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and dashboards shall load within a reasonable time for normal student feedback data volumes.
NFR-02	Security	Users shall access feedback records and application features permitted according to their assigned roles.
NFR-03	Usability	The application shall provide a simple and user-friendly interface accessible through web and mobile platforms.
NFR-04	Availability	The application shall be available through Zoho Creator cloud services with minimal downtime.
NFR-05	Maintainability	The application shall support easy modification of forms, workflows, reports, and business logic.
NFR-06	Auditability	Changes made to student, course, and feedback records shall be traceable through available application history and audit features.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Education / Event Edition
Development Environment	Zoho Creator Online Platform
Scripting Language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox
AI Assistant	Zia AI Assistant
Database	Zoho Creator Built-in Database
Other Zoho Services	Zoho Mail for notifications
Mobile Support	Zoho Creator Mobile Application (Android/iOS)
Client Hardware	Intel Core i3/i5 processor or above, 8 GB RAM, stable internet connection

3.5 Use Case Model

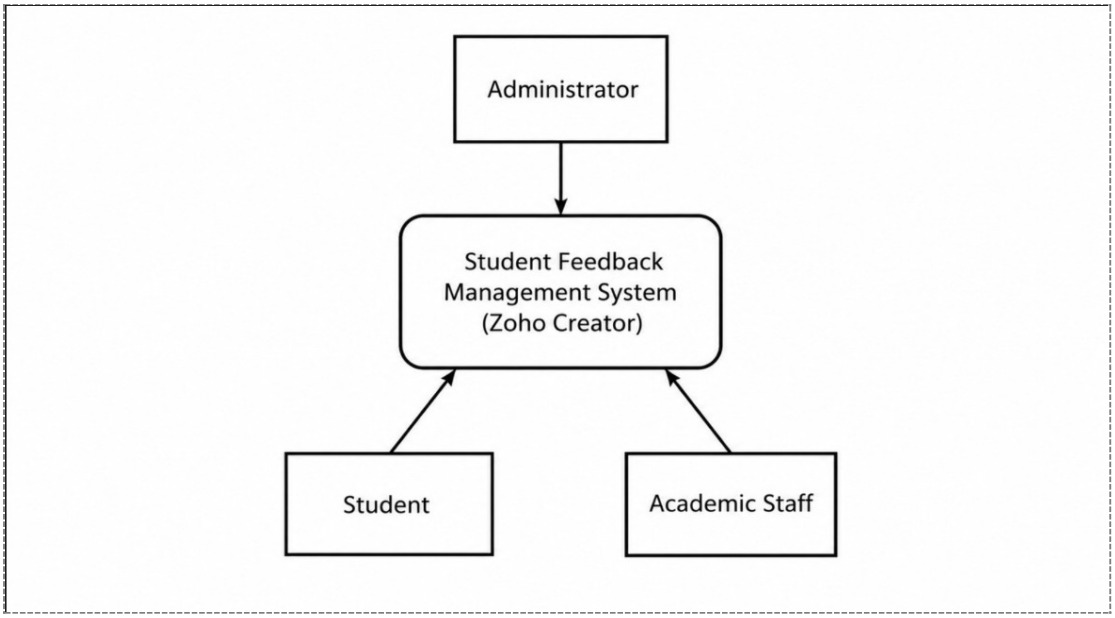


Figure 3.1: Use Case Diagram of Student Feedback Management System

Table 3.5: Use Case Description

Attribute	Details
Use Case ID	UC-01
Use Case Name	Student Feedback Submission
Actor(s)	Student
Pre-condition	The student must have access to the application, and the required student and course records must be available.
Main Flow	1. User opens Feedback Response Form. 2. Select the Student and Course details using lookup fields. 3. Enter feedback ratings, overall satisfaction. 4. System stores feedback response.
Alternate Flow	If mandatory fields are empty, the system displays validation errors.
Post-condition	The feedback response is created successfully.
Related FR	FR-03, FR-04

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

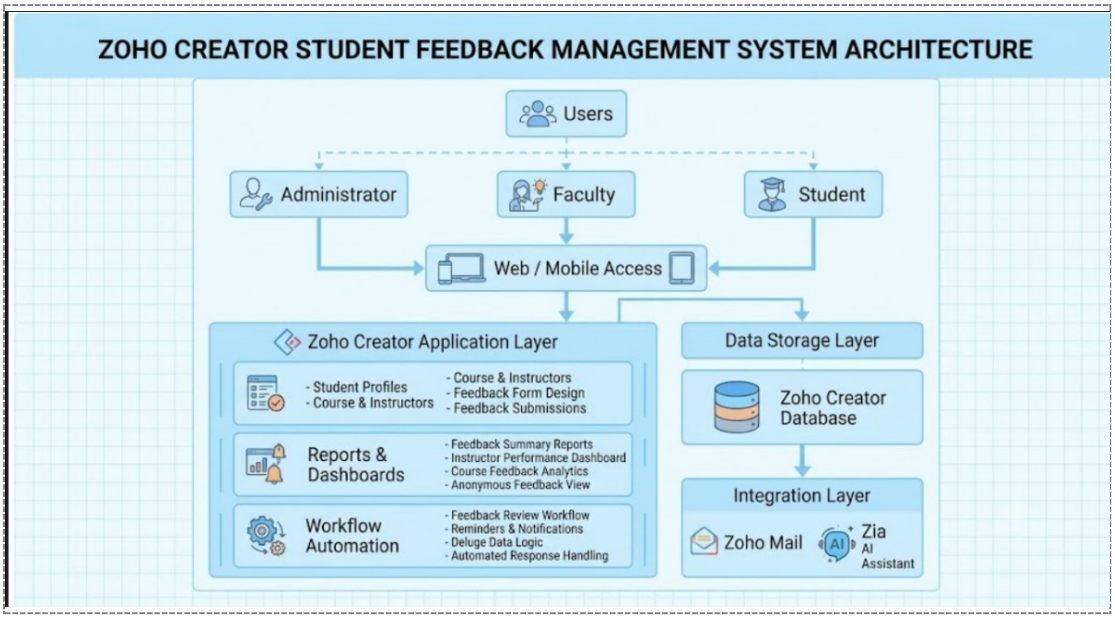


Figure 4.1: System Architecture of Student Feedback Management System

The Student Feedback Management System is designed using Zoho Creator, a cloud-based low-code application development platform. The system follows a layered architecture consisting of the User Layer, Application Layer, Business Logic Layer, and Database Layer. This architecture provides a structured approach to managing academic feedback information and automating application operations. The User Layer consists of three primary roles: Administrator, Student, and Academic Staff. The Application Layer includes the Student, Course, Feedback Response, and Semester Summary forms, along with reports and dashboards. The Business Logic Layer manages feedback validation, Deluge functions, workflow automation, email notifications, and Blueprint-based processes. The Database Layer stores student records, course information, feedback responses, and semester summaries using Zoho Creator's built-in database.

4.2 Data Model

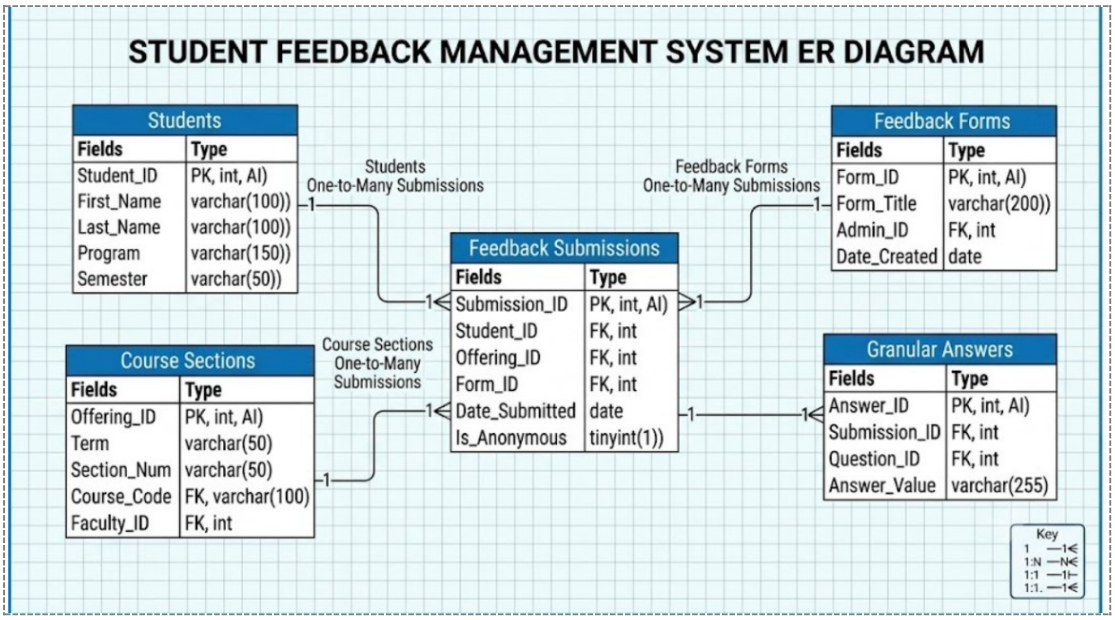


Figure 4.2: Entity Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Student Profiles	Student ID, Name, Register Number, Department, Semester, Email	Feedback Response	One-to-Many	Lookup Field
Course	Course ID, Course Name, Course Code, Instructor, Semester, Credits	Feedback Response	One-to-Many	Lookup Field
Feedback Response	Response ID, Student, Course, Content Rating, Delivery Rating, Clarity Rating, Engagement Rating, Overall Satisfaction	Student, Course	Many-to-One	Lookup Field
Semester Summary	Summary ID, Semester, Total Responses, Average Ratings, Trend Status	Feedback Response	Logical Association	Semester-wise Summary
Course Feedback Criteria(Subform)	Criteria Name, Description, Rating Scale	Course	One-to-Many	Subform

4.3 Form Design

Table 4.2: Field Specification – Student Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student ID	Student_ID	Auto Number	Yes	Prefix: STU-	System Generated
Student Name	Student_Name	Name	Yes	-	Student full name
Register Number	Register_Number	Single Line	Yes	Unique value	Student identification
Department	Department	Drop Down	Yes	AIML, CSE, IT	Academic department
Semester	Semester	Number	Yes	1-8	Current semester
Email	Email	Email	Yes	Valid email format	Student email
Phone Number	Phone_Number	Phone	No	10 digits	Contact number

Table 4.3: Field Specification – Course Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
CourseID	Course_ID	Single Line	Yes	Unique value	Course identification
CourseName	Course_Name	Single Line	Yes	-	Name of course
CourseCode	Course_Code	Single Line	Yes	Unique value	Academic course code
Credits	Credits	Number	Yes	Positive value	Course credit details
Instructor	Instructor	Single Line	Yes	-	Assigned course instructor

4.4 Process and Workflow Design

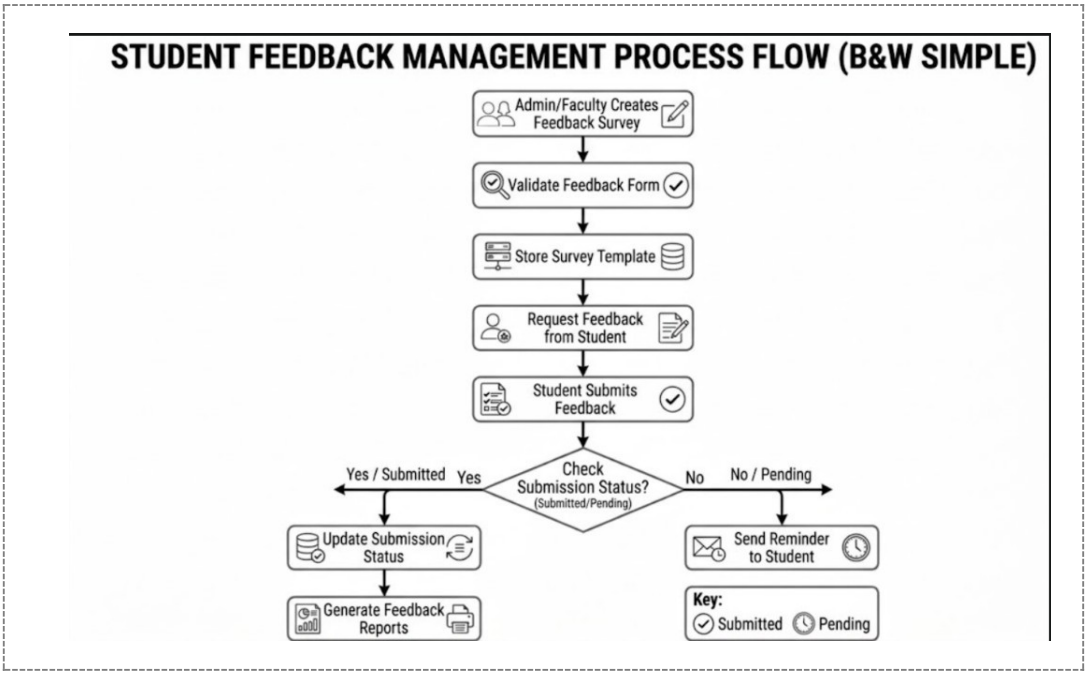


Figure 4.3: Workflow Process Flow for Student Feedback Management

The workflow starts when a student selects the respective course and enters feedback ratings. The system validates mandatory information before saving the response.

After feedback submission, the response is stored in the system, and a confirmation notification is generated.

Using Zoho Creator workflows and Deluge automation, the system validates feedback ratings and supports scheduled semester summary reporting.

Submitted feedback is reflected in reports and dashboards for monitoring course evaluation results and analysing academic feedback trends.

4.5 User Interface and Navigation Design

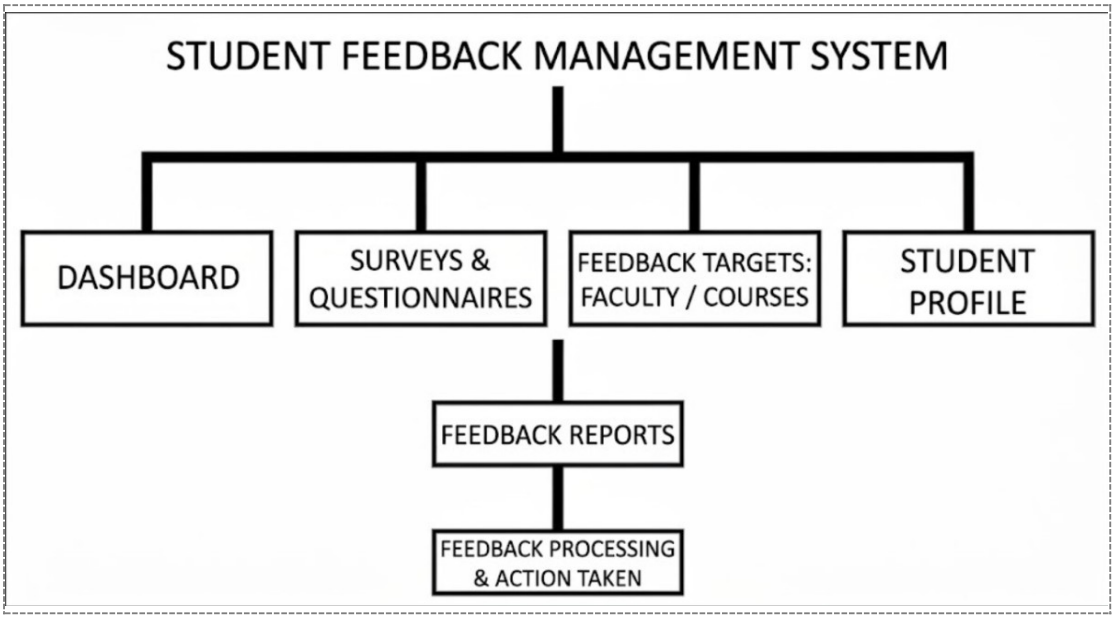


Figure 4.4: Student Feedback Management Dashboard

4.6 Access Control Design

Table 4.4: Role–Permission Matrix

Component	Administrator	Faculty	Student
Student Profiles Form	A E V D	V	V(Own)
Course Form	A E V D	V	V
Feedback Response Form	A E V D	V	A
Semester Summary Form	A E V D	V	—
All Responses Report	V	V	V(Own)
Semester Summary Report	V	V	—
Dashboard	V	V	—
Workflow Settings	A E D	—	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The Student Feedback Management System was developed using Zoho Creator, a low-code application development platform. The application was created under the name Student Feedback Collection and Course Evaluation to simplify academic feedback collection and analysis. The application consists of four main modules: Student, Course, Feedback Response, and Semester Summary. The navigation menu was organised to provide access to forms, reports, dashboards, and workflows. Zoho Creator's visual development tools were used to customise the application according to the project requirements.

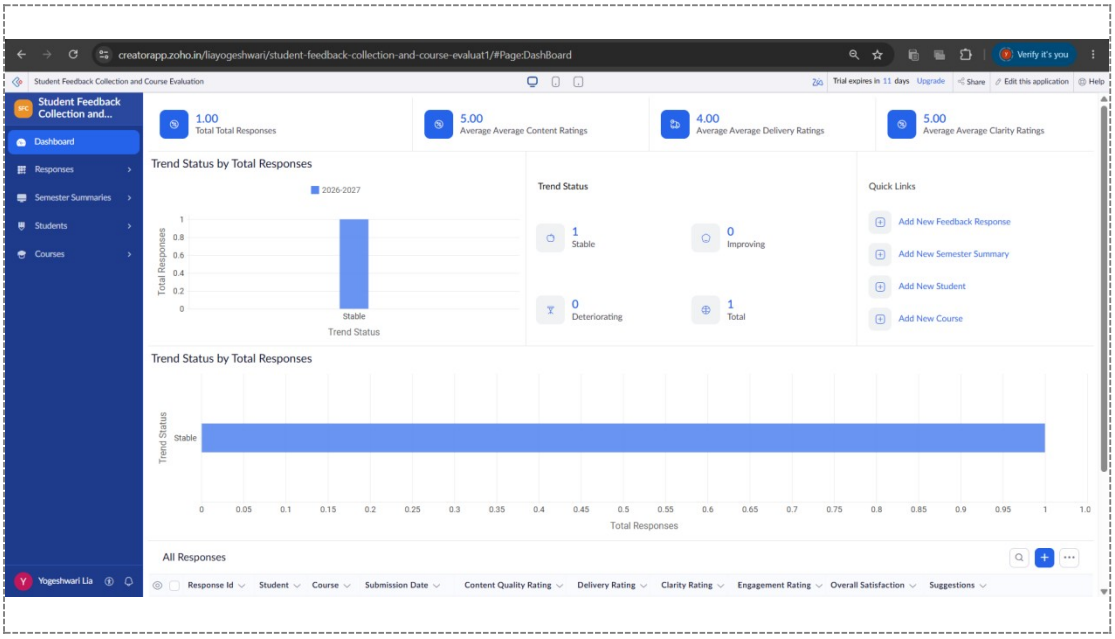


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

Four forms were developed to capture and manage academic feedback information. The Student Form maintains student details, while the Course Form stores course information such as course name, code, instructor, semester, and credits. The Feedback Response Form allows students to evaluate courses based on Content Quality, Delivery, Clarity, Engagement, and Overall Satisfaction. It also provides fields for suggestions and anonymity preference. Lookup fields connect feedback responses with the corresponding Student and Course records.

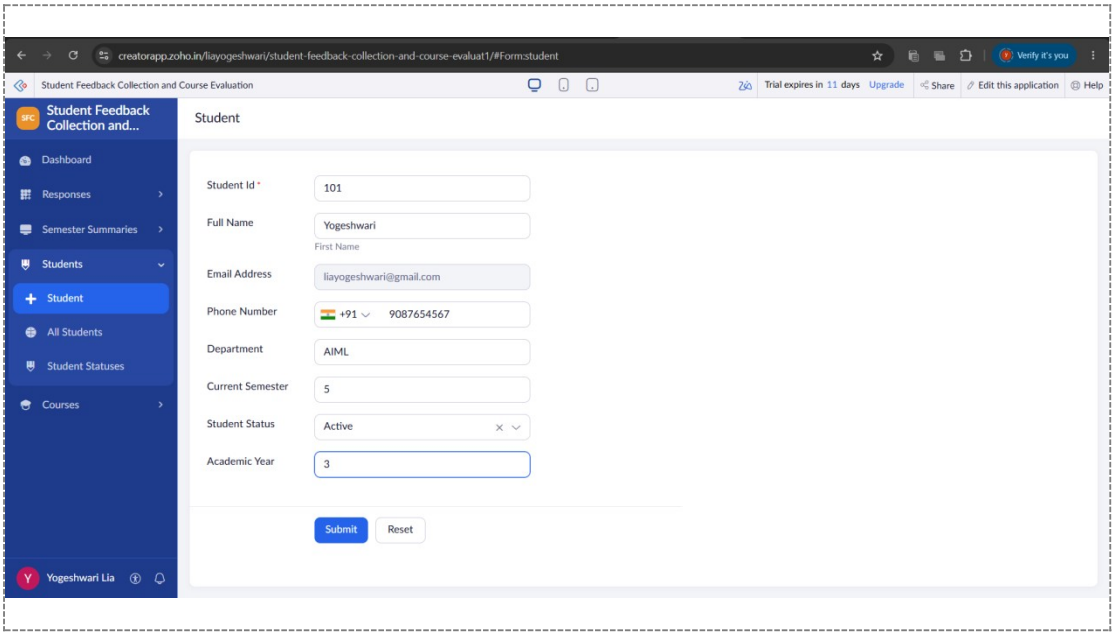


Figure 5.2: Student Form Builder

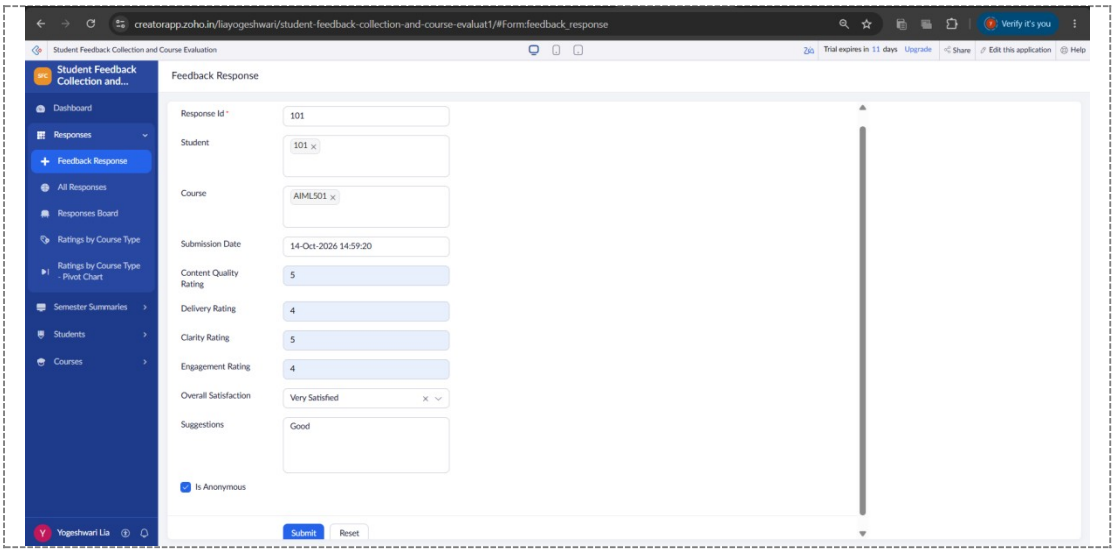


Figure 5.3: Feedback Response Builder

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Students	List Report	Student	Sort by Student ID	View Details	Administrator
All Courses	List Report	Course	Sort by Course Code	View Details	Administrator, Academic Staff
All Responses	List Report	Feedback Response	Sort by Submission Date	View Feedback	Administrator, Academic Staff
Responses Board	Kanban Report	Feedback Response	Group by overall Satisfaction	View Response	Administrator, Academic Staff
Ratings by Course Type-Pivot Chart	Pivot Chart	Feedback Response	Group by Course Type and Ratings	Analyse Ratings	Administrator, Academic Staff

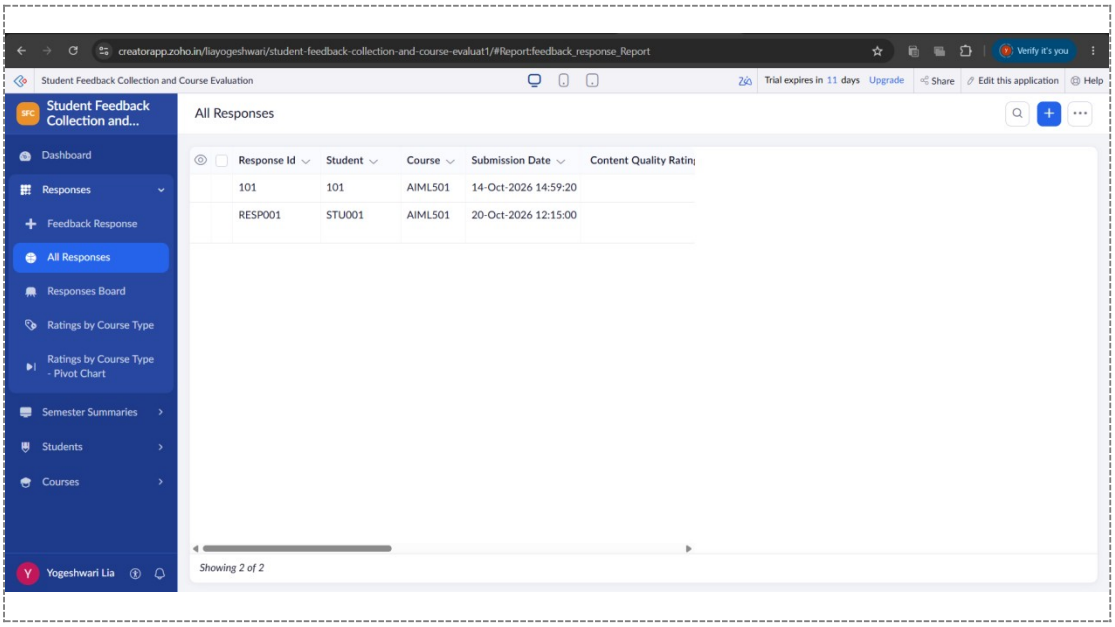


Figure 5.4: Feedback Reports and Analytics

5.4 Pages and Dashboards

A dashboard page was developed using Zoho Creator's Page Builder to provide a visual overview of the collected feedback. The dashboard displays key performance indicators, including total feedback responses and average ratings for Content, Delivery, and Clarity. A chart representing Trend Status helps academic staff review semester-wise feedback patterns.

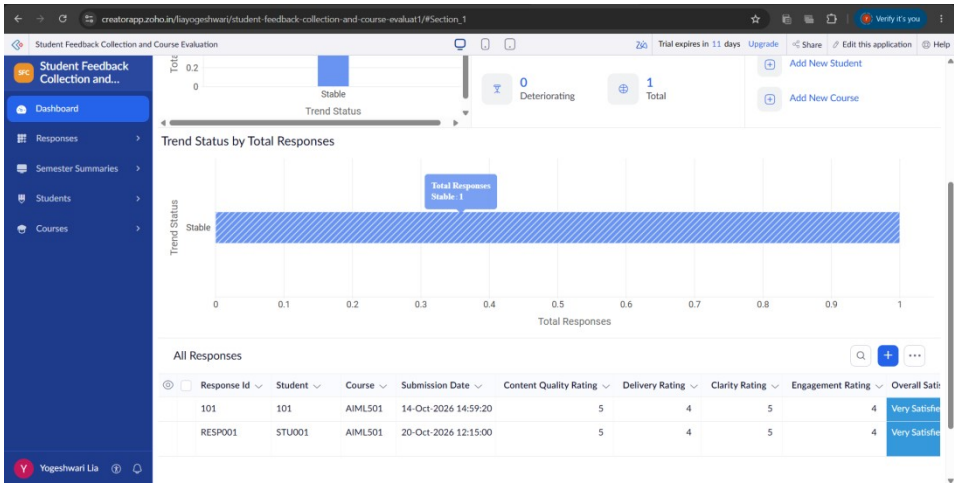


Figure 5.5: Student Feedback Management Dashboard

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Student Form → On Load	Automatically fill Student Email using the logged-in user's email address	zoho.loginuserid, input assignment, disable field
DS-02	Feedback Response Form → On Validate	Validate feedback ratings before form submission	If condition, logical operators, validation
DS-03	Functions → Get_Course_Feedback_Count	Count feedback responses for a selected course	Criteria-based record fetch, loop, counter
DS-04	Monthly Semester Summary → Scheduled Workflow	Retrieve semester summary records for periodic reporting	Criteria-based fetch, collections, iteration

Code 5.1: Auto-fill Student Email Script

```
input.email = zoho.loginuserid;
disable email;
```

This script automatically fills the Student Email field using the logged-in user's email address and disables manual editing of that field.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Reference
Auto-fill Student Email	Form Workflow	Student Form → On Load	Student opens the form	Automatically populate Student Email	DS-01
Validate Feedback Ratings	Form Workflow	Before Feedback Submission	Ratings must be within the valid range (1–5)	Validate ratings and prevent invalid submission	DS-02
Course Feedback Collection	Blueprint	Feedback process transition	Feedback progresses through the defined stages	Manage feedback collection stages	—
Monthly Semester Summary Report	Schedule Workflow	Monthly Schedule	Scheduled execution date is reached	Retrieve semester summary records for reporting	DS-04

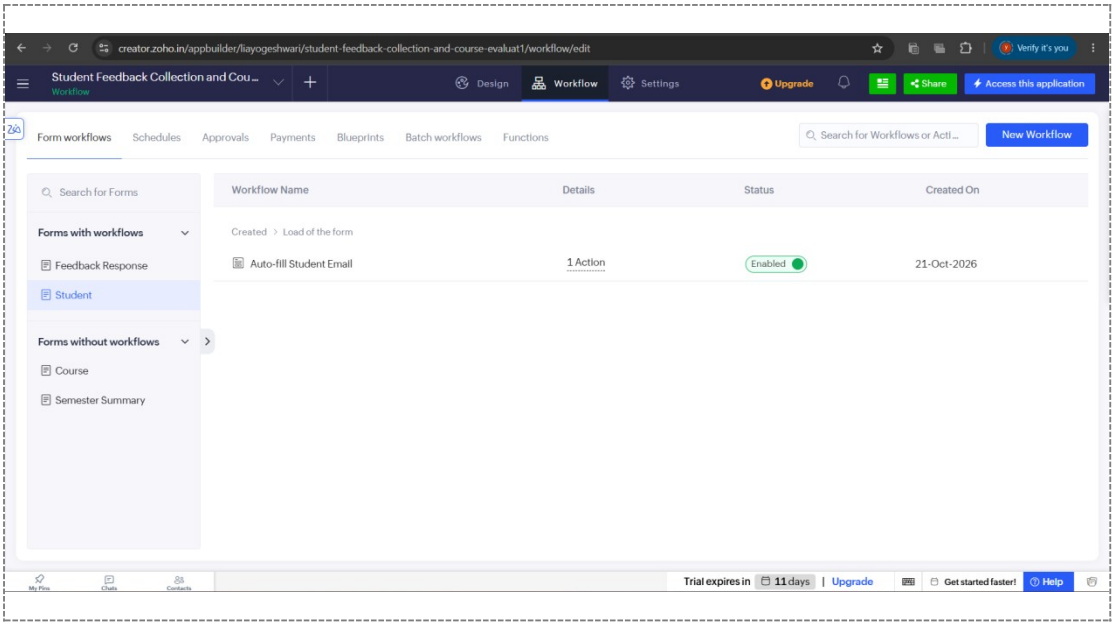


Figure 5.6: Auto-fill Student Email Workflow Blueprint

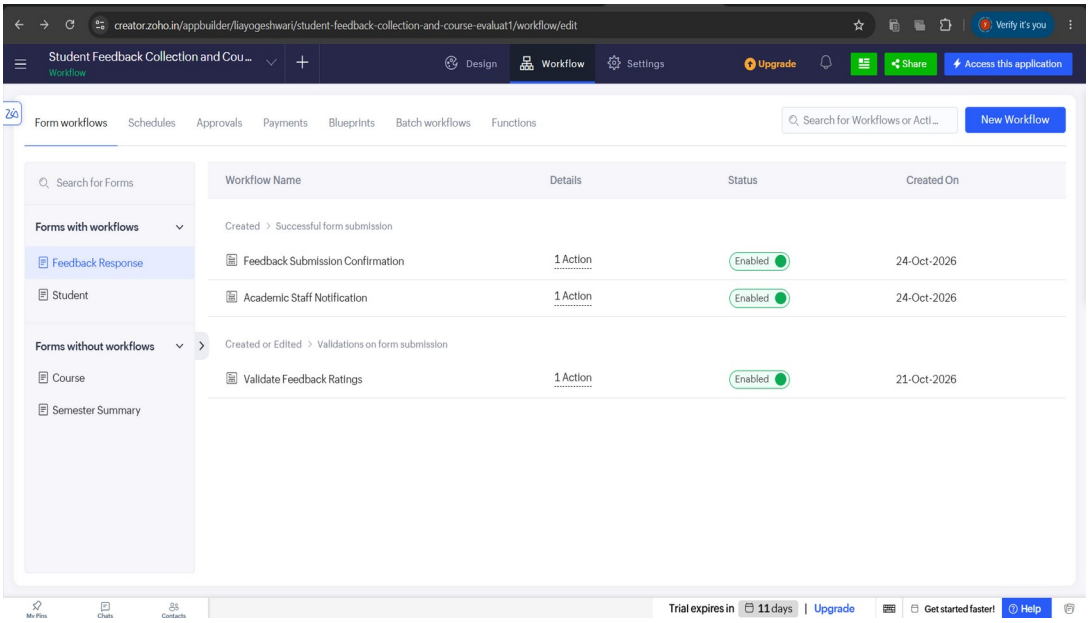


Figure 5.7: Validate Feedback Ratings Workflow

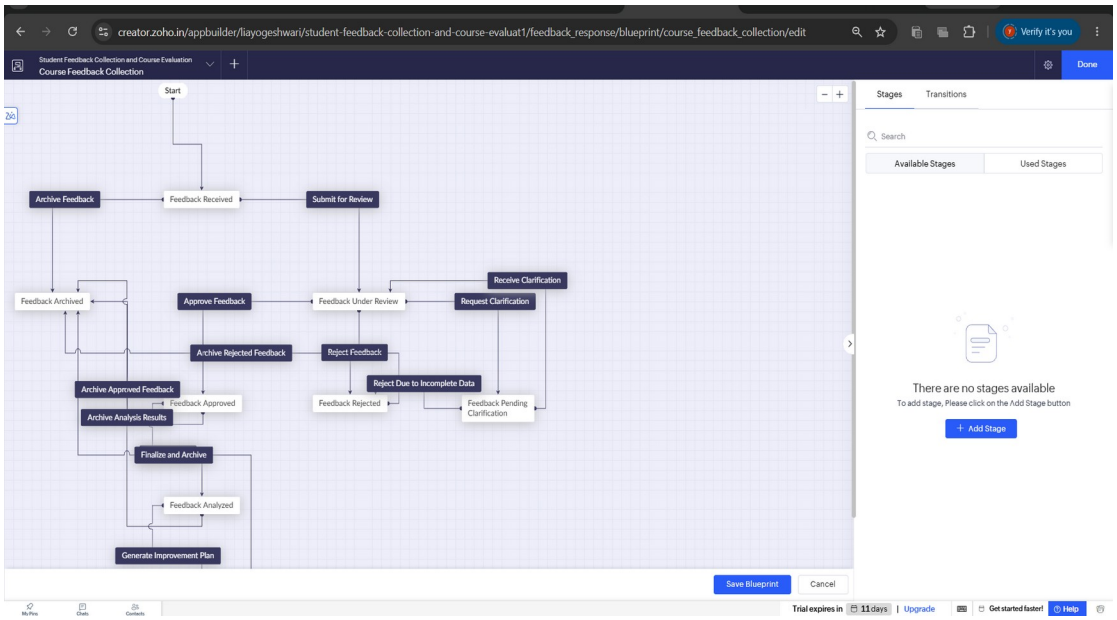


Figure 5.8: Course feedback Collection Blueprint

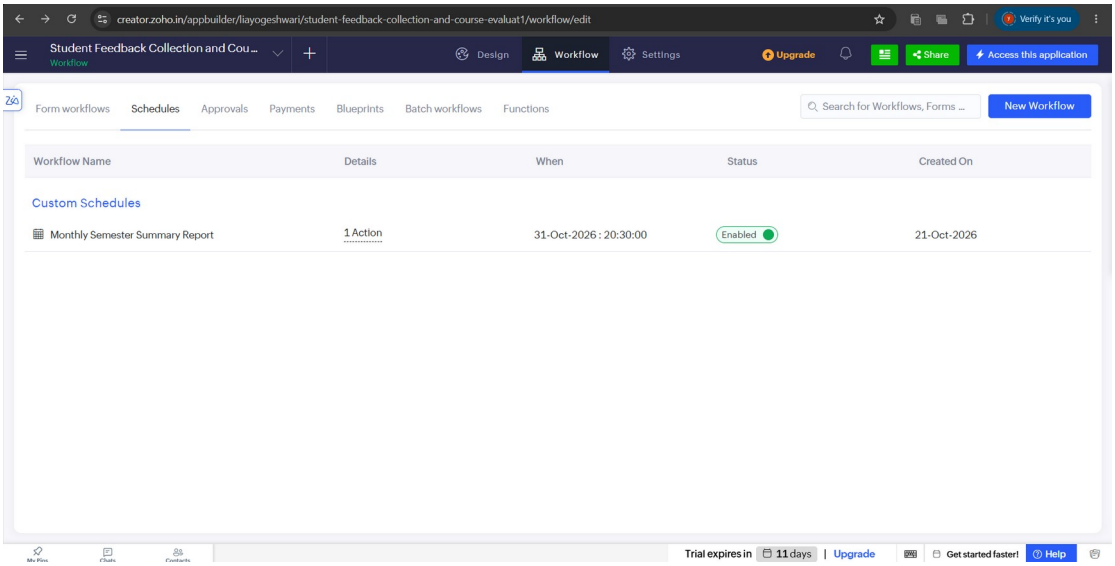


Figure 5.9: Monthly Semester Summary Scheduled Workflow

5.7 Integrations and AI-Assisted

Table 5.4: Integrations Implemented

Integration	Service	Method	Authentication	Purpose
Email Notification	Zoho Mail	Built-in Email Function	Zoho Authentication	Send feedback submission confirmations and academic staff notifications
AI Assistance	Zia AI Assistant	AI-Assisted Development	Zoho Account	Assist in application creation and configuration
Analytics	Zoho Creator Charts	Built-in Reports and Dashboard	Zoho Authentication	Visual analysis

Code 5.3: **Student Email Auto-fill Script**

```
// Automatically fetch the logged-in user's email

input.email = zoho.loginuserid;

// Prevent manual modification of the email field
```

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

The Student Feedback Management System was evaluated using functional testing to examine the behaviour of forms, reports, workflows, and dashboard components. The testing strategy focused on verifying data entry, mandatory field validation, feedback submission, and automated operations.

6.2 Test Cases and Results

Table 6.1: Test Cases

Test ID	Test Description	Expected Result	Status
TC-01	Submit Student Form with valid details	Student record created successfully	Pass
TC-02	Submit form without mandatory fields	Validation error displayed	Pass
TC-03	Add new course details	course record stored successfully	Pass
TC-04	Submit feedback using Student and Course lookups	Feedback Response created successfully	Pass
TC-05	Open Student Form	Logged-in user's email automatically populated	Pass
TC-06	Update Enter feedback rating outside the permitted range	Invalid submission prevented	Pass
TC-07	Submit valid feedback ratings	Feedback Stored Successfully	Pass
TC-08	View submitted feedback in All Responses	Feedback records displayed correctly	Pass
TC-09	Academic Staff views feedback reports	Permitted feedback information displayed	Pass
TC-10	Unauthorized user accesses administrator features	Access restricted	Pass
TC-11	Generate course feedback report	Report displayed correctly	Pass
TC-12	Open Dashboard	Feedback KPI values displayed	Pass
TC-13	View Semester Summary	Summary information displayed correctly	Pass
TC-14	Delete record with permission	Record deleted successfully	Pass
TC-15	Login with valid credentials	User logged in successfully	Pass

6.3 Debugging Deluge Scripts

Validation mechanisms were incorporated into the application to improve data accuracy and prevent incorrect feedback submissions. Zoho Creator's built-in field validation and form configuration features support consistent data entry and help reduce incomplete or invalid records.

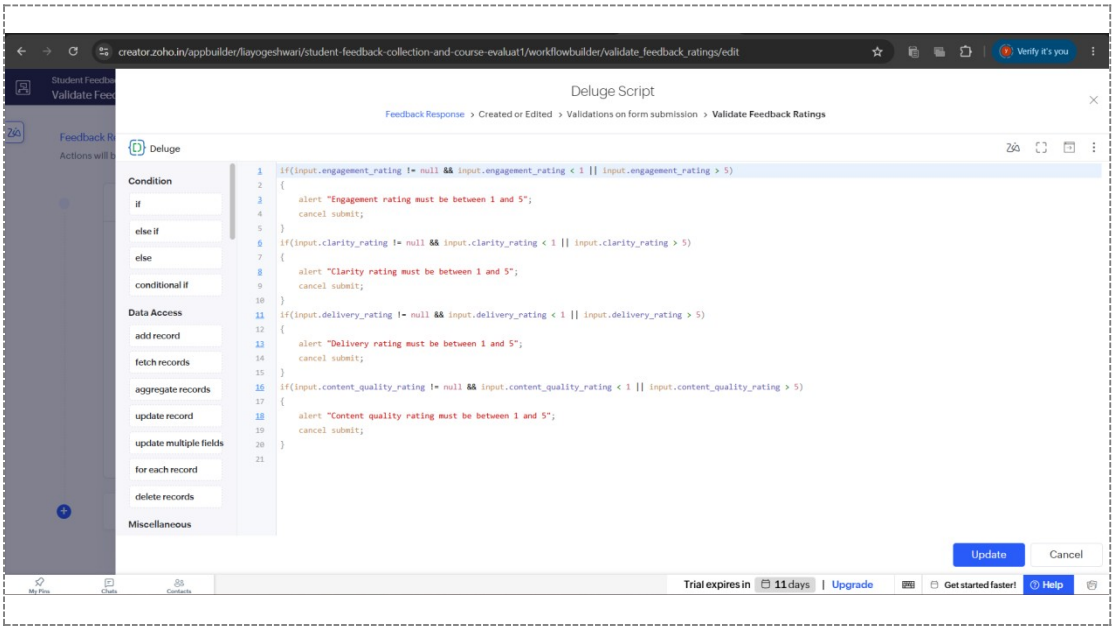


Figure 6.1: Feedback Rating Validation Error Message

6.4 Issues Identified and Resolutions

Table 6.2: Issues Identified and Resolutions

Bug ID	Issues Identified	Root Cause	Resolution	Status
IS-01	Student Email required automatic population	Email auto-fill workflow was not configured	Added an On Load Deluge script using zoho.loginuserid	To Verify
IS-02	Feedback ratings required validation	Rating validation rules were not configured	Added validation workflow to check feedback ratings	To Verify

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Application Access and Deployment Status

The Student Feedback Management System was developed using Zoho Creator, a cloud-based low-code application development platform. The application, named Student Feedback Collection and Course Evaluation, is maintained within the Zoho Creator environment.

The application provides access to Student, Course, Feedback Response, and Semester Summary modules through its navigation interface. Forms, reports, dashboards, and workflows are organised within the application to support academic feedback management.

The current implementation is intended for academic demonstration and evaluation. Deployment across separate development, testing, and production environments was not performed.

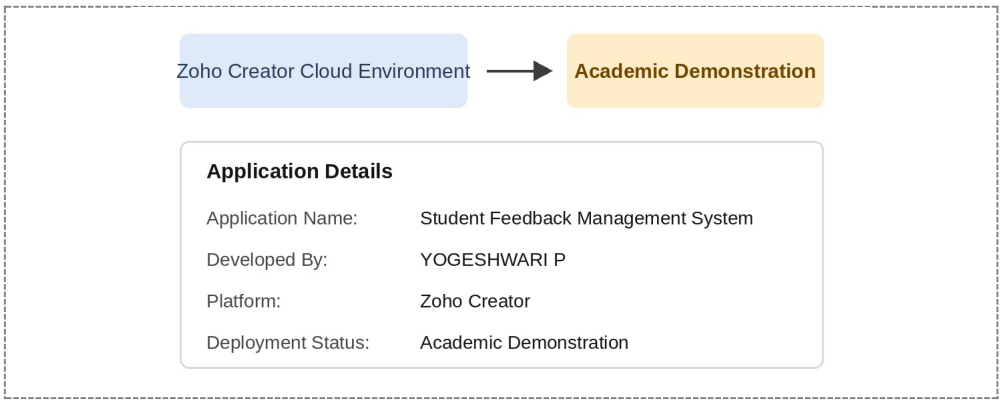


Figure 7.1: Environment & Deployment

7.2 Performance Considerations

Performance considerations were taken into account while designing the application to support efficient feedback collection and retrieval.

Zoho Creator's built-in database and report components were used to organise records and display feedback information. Lookup fields connect related student, course, and feedback records, reducing unnecessary duplication.

The dashboard presents important feedback metrics, including total responses and average ratings, in a centralised interface. Reports provide structured access to submitted feedback and semester summary information.

7.3 Security Features

The application uses Zoho Creator's built-in authentication and access management

capabilities to support secure application usage. Role-based access is designed around three user categories: Administrator, Academic Staff, and Student. These roles define the intended responsibilities and access requirements for different application components.

The Student Email field uses the logged-in user's email address through the zoho.loginuserid Deluge function. Mandatory field settings and feedback validation workflows help maintain data accuracy.

Table 7.2: Security Features

Security Features	Purpose
User Authentication	Identify application users
Role-Based Access	Organise access permissions
Email Auto-fill	Populate logged-in student email
Field Validation	Reduce invalid data entry

7.4 Governance and Administration

The Administrator is responsible for maintaining student and course information, reviewing submitted feedback, and managing application components. Academic Staff can use the permitted reports and dashboard components to review feedback results and monitor course evaluation trends. Students are intended to submit feedback through the Feedback Response Form.

Application maintenance includes reviewing form configurations, workflow settings, report structures, and user permissions. Changes to the application should be checked before use to maintain data consistency and reliable operation.

The system demonstrates how Zoho Creator's built-in administration and access management features can support the governance of an academic feedback application.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The Student Feedback Management System was developed using Zoho Creator to simplify academic feedback collection and course evaluation. The application includes Student, Course, Feedback Response, and Semester Summary forms for managing academic information.

Scenario:

Step 1: Administrator creates student and course records with the required academic details.

Step 2: Student selects the respective course and submits feedback ratings and suggestions.

Step 3: Dashboard displays total feedback responses, average ratings, and course evaluation trends.

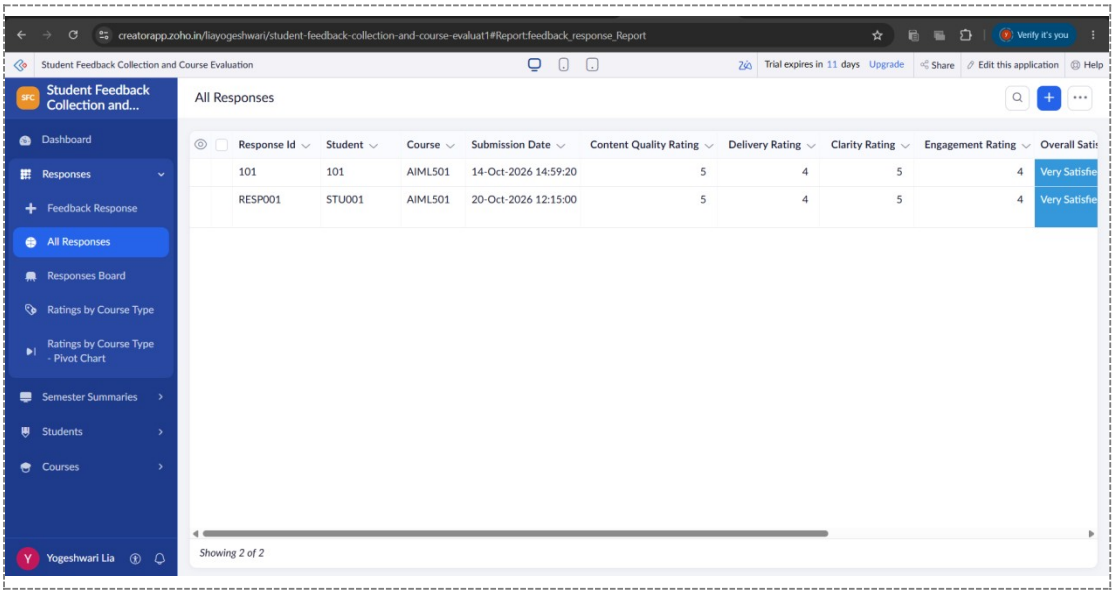


Figure 8.1: Student Feedback Management Application Response

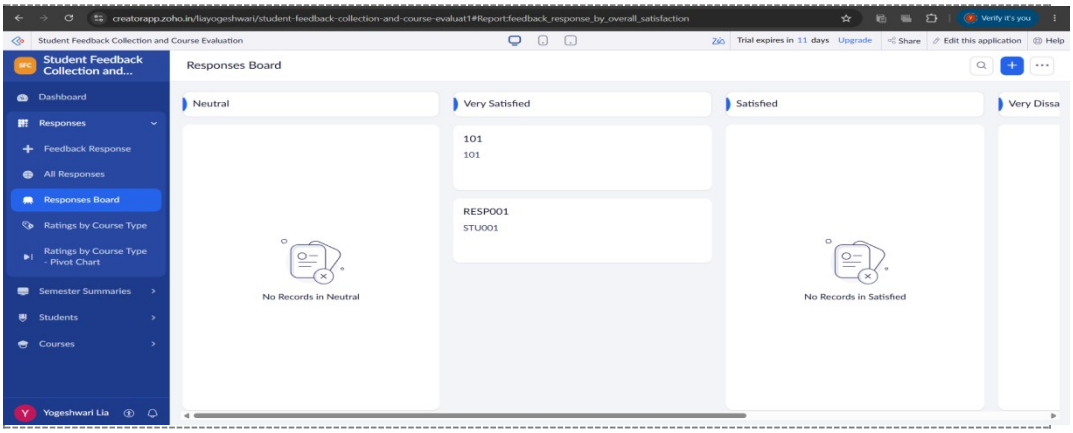


Figure 8.2: Student Response Report

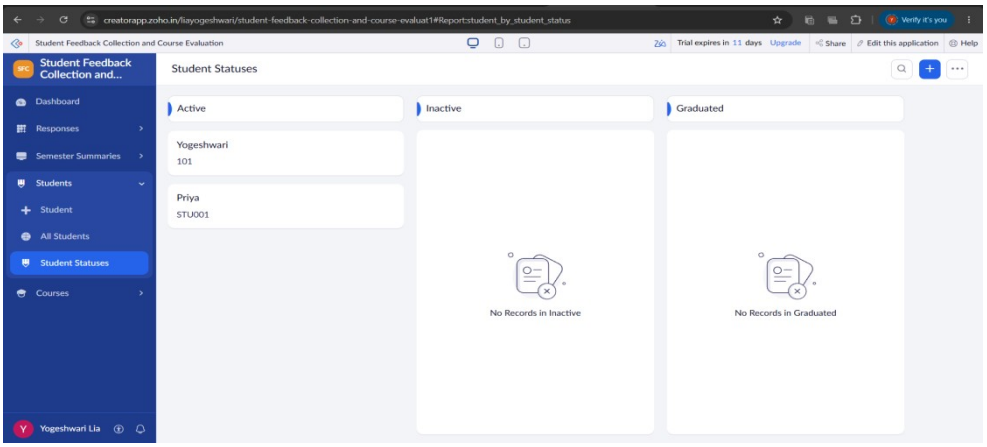


Figure 8.3: Student Report

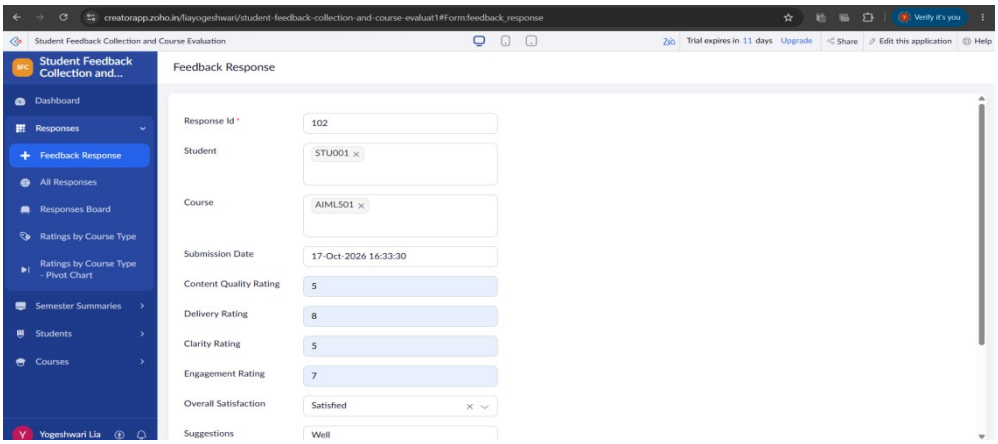


Figure 8.4: Feedback Report Form

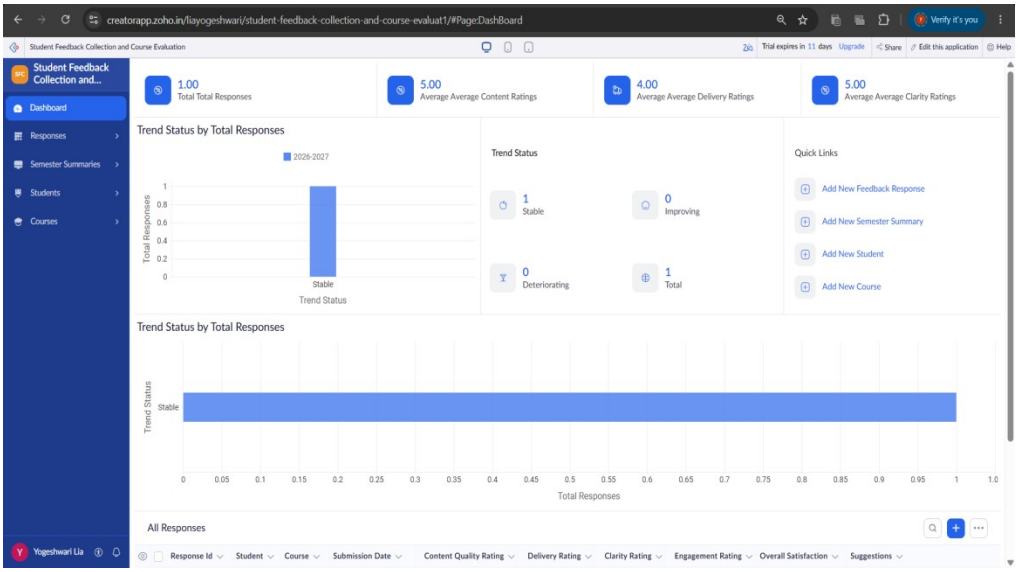


Figure 8.5: Dashboard

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective	Implementation	Outcome
Manage student feedback	Zoho Creator Forms	Digital feedback management
Evaluate course performance	Feedback Response and semester Summary Forms	Structured course evaluation
Generate insights	Dashboards and Reports	Better academic feedback analysis

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Feedback Collection	Manual feedback forms	Online feedback submission	Saves time
Course Evaluation	Manual analysis	Rating-based evaluation	Better accuracy
Reports	Manual preparation	Built-in reports and charts	Faster analysis
Data Access	Limited	Role-based access	Organized and controlled access

8.4 Limitations of the Low-Code Approach

The application has certain limitations, such as restricted UI customisation, dependency on Zoho Creator features, and limited support for complex feedback analysis. These limitations can be addressed through additional Deluge scripting, external integrations, and future pro-code enhancements.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The Student Feedback Management System uses no-code features to create forms, reports, and dashboards. Low-code development was implemented through Deluge scripting, workflows, and Blueprints for feedback validation and automation. Future enhancements involving advanced feedback analysis, AI-based sentiment analysis, or external API integrations can be developed using pro-code technologies.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Student Feedback Management System was developed using Zoho Creator to simplify academic feedback collection and course evaluation. The application provides digital forms for managing student details, course information, feedback responses, and semester summaries.

9.2 Future Enhancements

• Integration with college ERP and student information systems.
• Advanced analytics to identify course performance comments.
• External API integration for connecting additional academic services.

9.3 Learning Outcomes and Career Relevance

The project also demonstrated the relationship between no-code, low-code, and pro-code development approaches. These skills are relevant to careers in software development, application engineering, business process automation, and AI-assisted application.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Yes	Student Feedback Collection and Course Evaluation Application	5.1	Fig 5.1
II	Forms with ≥ 8 distinct field types	Yes	Student, Course, Feedback Response and Semester Summary Forms	5.2	Fig 5.2-5.3
II	Lookup field (relationship between forms)	Yes	Feedback Response Form – Student and Course Lookup Fields	4.2,5.2	Fig 4.2,5.3
II	Subform (parent-child data)	Yes	Course Form – Course Feedback Criteria Subform	5.2	Table 4.1
II	Field validations / layout rules	Yes	Feedback Response – Validate Feedback Ratings Workflow	5.5,6.2	Code C.2, Fig. 6.1
II	List report with filter, grouping, custom action	Yes	All Responses Report	5.3	Fig 5.4
II	Kanban / Calendar / Timeline / Map report	Yes	Responses Board (Kanban Report)	5.3	Fig 8.2
II	Pivot table / chart report	Yes	Ratings by Course Type – Pivot Chart	5.3	Table 5.1
II	Page / dashboard built with UI Builder	Yes	Student Feedback Dashboard	5.4	Fig 5.5
II	Deluge – form events	Yes	Auto-fill Student Email and Validate Feedback Ratings	5.5	Code C.1,C.2
II	Deluge – functions, fetch, loops, collections	Yes	Default Namespace – Three Standalone Deluge Functions	5.5	Code C.2,C.8

Unit	Zoho Creator Feature	Done	Where in Application	Report Section	Figure / Code No.
II	Email / in-app notifications	Yes	Feedback Submission Confirmation and Academic Staff Notification	5.6	Code C.3, C.4
II	Form workflow	Yes	Four Feedback Response Form Workflows	5.6	Fig 5.7
II	Scheduled workflow	Yes	Monthly Semester Summary Report	5.6	Code C.5
II	Approval process	No	Not implemented	-	-
II	Blueprint	Yes	Course Feedback Collection and Semester Performance Review	5.6	Fig. 5.8
II	Batch / report workflow	Yes	Monthly Semester Summary Report (scheduled summary processing)	5.6	Code C.5
II	Testing & debugging evidence	Yes	Feedback Rating Validation and Deluge Script Testing	6.2,6.3	Fig 6.1
III	Integration with a Zoho service	No	Not implemented	-	-
III	Connection / REST API call (invokeurl)	No	Not implemented	-	-
III	Webhook or Creator REST API	No	Not implemented	-	-
III	Users, roles and permission profiles	Yes	Administrator, Student and Academic Staff Permission Profiles	4.6,7.3	Table 4.4
III	Record-level access criteria	Yes	Application Settings – User Permissions and Record Sharing	4.6,7.3	Table 4.4

Unit	Zoho Creator Feature	Done	Where in Application	Report Section	Figure / Code No.
III	Zia / AI feature	Yes	Zia Assistant – Application Development Assistance	5.7	-
III	Analytics (charts / Zoho Analytics)	Yes	Dashboard – Trend Status Chart and Ratings by Course Type Pivot Chart	5.3,5.4	Fig 5.5
IV	Performance optimisation applied & measured	No	No measured performance benchmark recorded	-	-
IV	Deployment across environments	No	Single Zoho Creator Cloud Environment	-	-
IV	Security controls (encryption, OAuth, permissions)	Yes	Application Settings – Authentication and Permission Profiles	7.3	Table 7.2,7.3
IV	Governance (audit trail, administration)	Yes	Application Administration	7.4	-
V	Pro-code extension (Catalyst/widgets/custom functions)	No	Future Enhancement Only	-	-
—	Mobile app / portal access	Yes	Zoho Creator Mobile Access	5.7	-

Total features implemented: 26 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Chapter 1 – Introduction; Chapter 2 – Literature Survey and System Study; Chapter 3 – Requirement Analysis	3 – High	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Chapter 4 – System Design; Chapter 5 (Sections 5.1–5.6) – Application Setup, Forms, Reports, Dashboard, Deluge Scripting and Workflows; Chapter 6 – Testing and Debugging	3 – High	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Section 4.6 – User Roles and Access Design; Section 5.7 – Integrations and Additional Features; Section 7.3 – Security and Access Control	2 – Medium	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Chapter 7 – Deployment, Security and Governance (Sections 7.1–7.4)	2 – Medium	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Section 5.7 – Integrations and Additional Features; Sections 8.4–8.5 – Results and Discussion; Section 9.3 – Future Enhancements	3 – High	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Auto-fill Student Email

```
Script ID : DS-01

// Location : Student Form → On Load

// Purpose : Automatically fill the Student Email field
//           using the logged-in user's email address.

input.email = zoho.loginuserid;

disable email;
```

C.2 Validate Feedback Ratings

```
// Script ID : DS-02
// Location : Feedback Response Form → On Validate
// Purpose : Validate feedback ratings before form submission

if(input.Content_Rating < 1 || input.Content_Rating > 5 ||
   input.Delivery_Rating < 1 || input.Delivery_Rating > 5 ||
   input.Clarity_Rating < 1 || input.Clarity_Rating > 5)
{
   alert "Please enter feedback ratings between 1 and 5."
   cancel submit;
}
```

C.3 Fetch Course Feedback Details

```
// Script ID : DS-03
// Location : Functions → Get_Course_Feedback_Count
// Purpose : Fetch and count feedback responses for a selected course

feedback_count = 0;

feedback_records = Feedback_Response[Course == course_id];

for each record in feedback_records
{
   feedback_count = feedback_count + 1;
}

return feedback_count;
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open the Student Feedback Collection and Course Evaluation application using a web browser or Zoho Creator mobile application.
2. Enter the registered username and password to log in.
3. After successful login, access the application according to the assigned user role.
4. Users can access Student, Course, Feedback Response, Semester Summary, reports and the dashboard from the application menu, subject to their permissions.

D.2 Instructions for Administrator

1. Login using administrator credentials.
2. Manage student profiles and course information.
3. Create, edit, and manage records in the Student, Course, Feedback Response and Semester Summary forms.
4. Monitor submitted feedback through the All Response report.
5. Manage user roles, permission profiles and application settings.

D.3 Instructions for Faculty

1. Login using faculty credentials.
2. View course details through the All Course report.
3. Monitor course feedback ratings and overall satisfaction levels.
4. View semester-wise feedback summaries.
5. Analyse feedback trends using the dashboard.
6. Check feedback submission notifications.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	YOGESHWARI P	Register No.	2116241501254
Project Title	Student Feedback Management System	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name:

Signature of External Examiner

Name:

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	YOGESHWARI P	Register No.	2116241501254
Department / Section	AIML	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: Name: